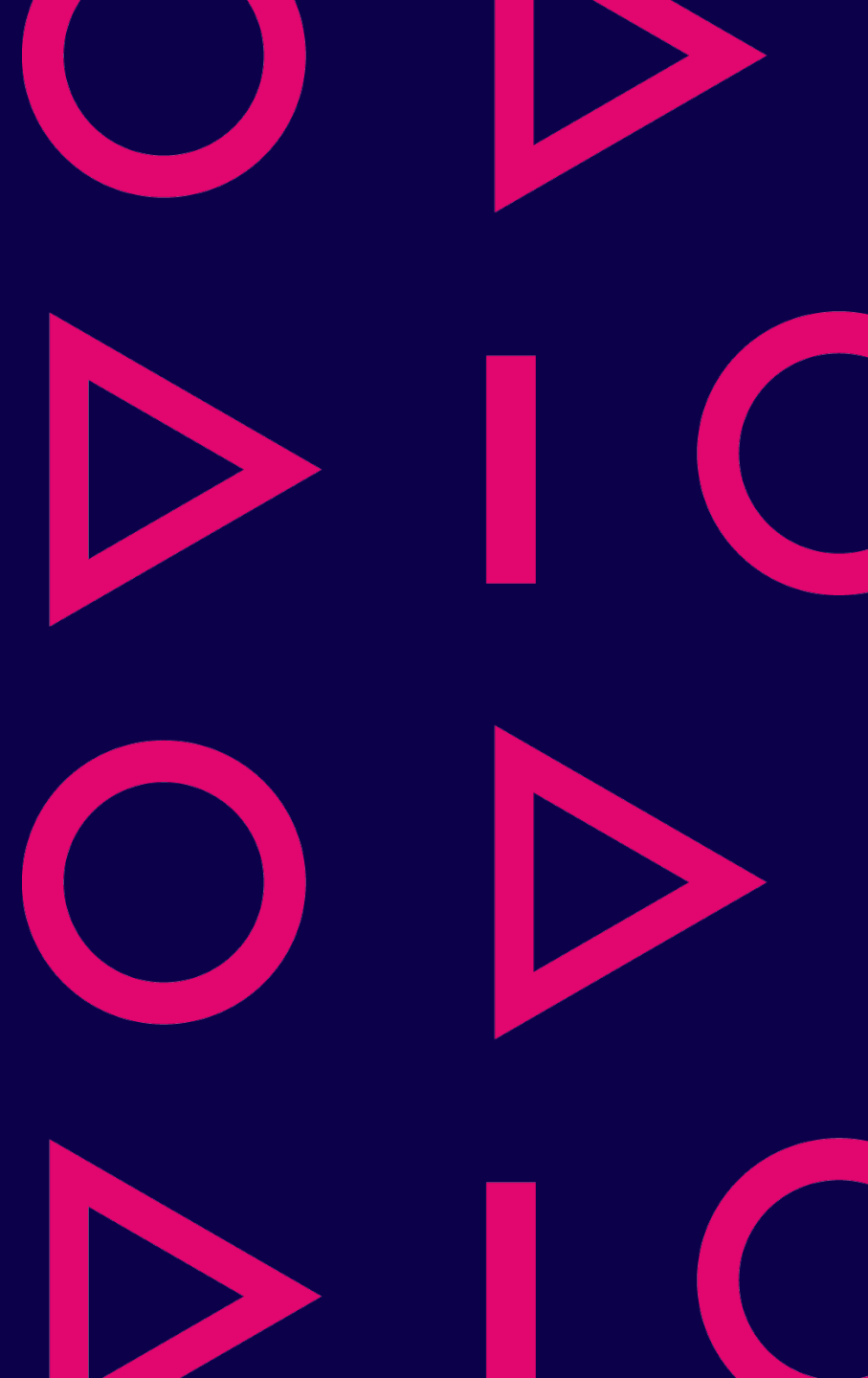


Basics of automation technology

Samppa Alanen

Ref. Teppo Flyktman



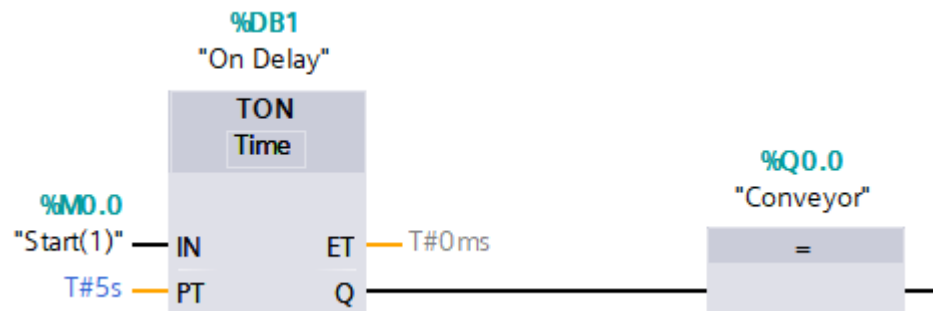
Timers

- On Delay
 - Delays signal/memory bit change from state 0 to 1
- Off Delay
 - Delays signal/memory bit change from state 1 to 0
- Pulse
 - Sets a memory bit/signal to 1 for the time specified by the pulse timer
- Not all logics necessarily have all types of timers.
- See Help for more details (activate the program block of interest and press F1). Works with programming tools for most logics and more

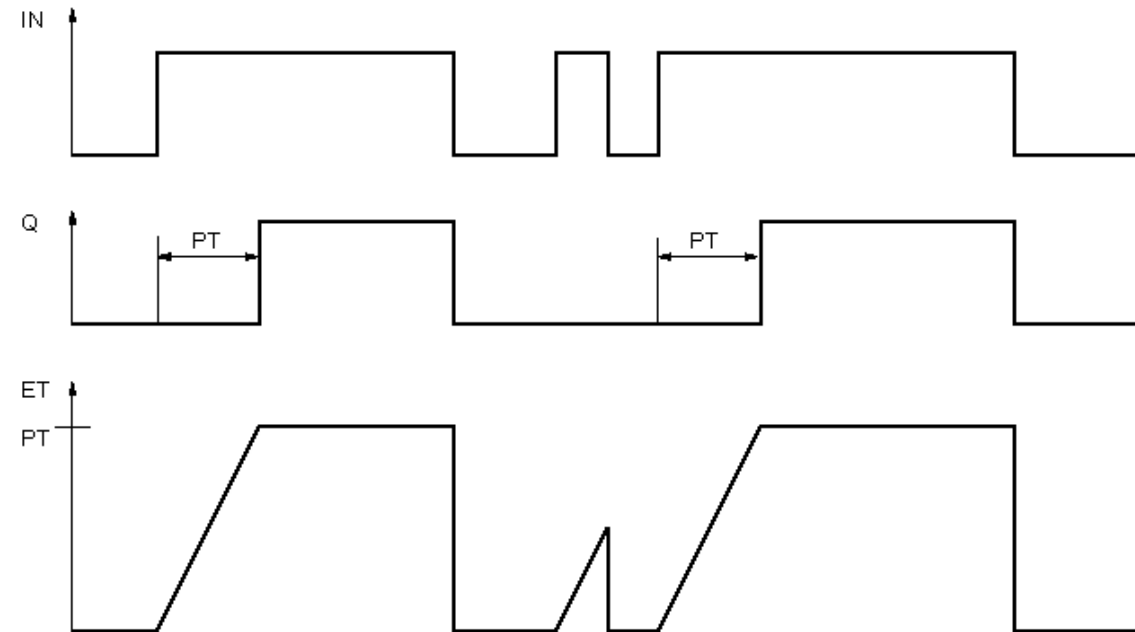
Timers

Siemens TIA Portal

On Delay



The following figure shows the pulse timing diagram of the "Generate on-delay" instruction:

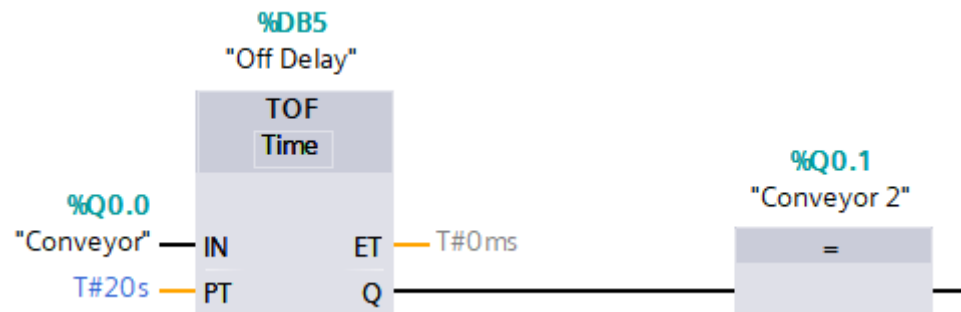


When the memory slot M0.0 changes to mode 1, a five-second on delay is triggered, after which the output Q0.0 is turned on. If the memory slot M0.0 changes to 0, the timer and its output will be reset. From the output of the timer "ET", the current value of the timer can be read.

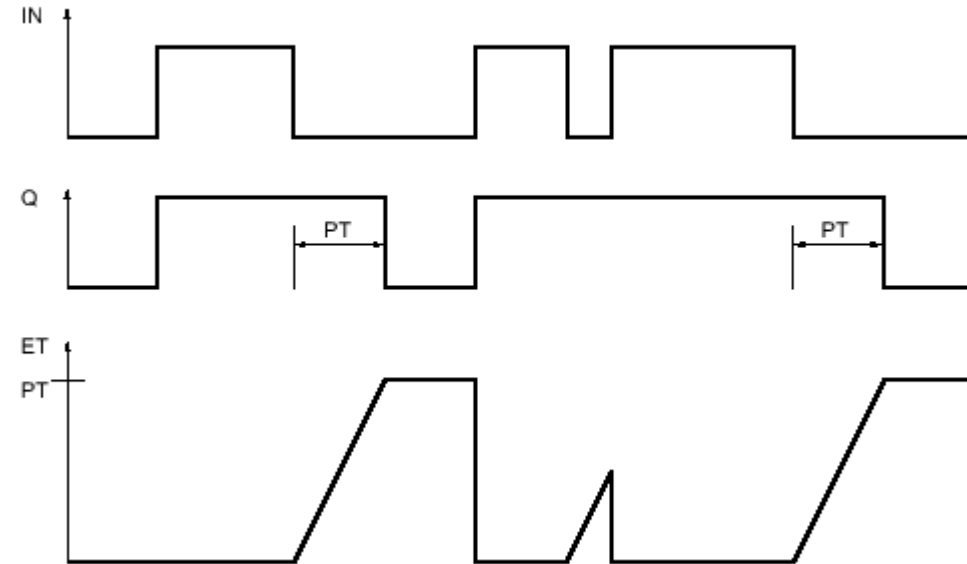
Timers

Siemens TIA Portal

Off Delay



The following figure shows the pulse timing diagram of the "Generate off-delay" instruction:

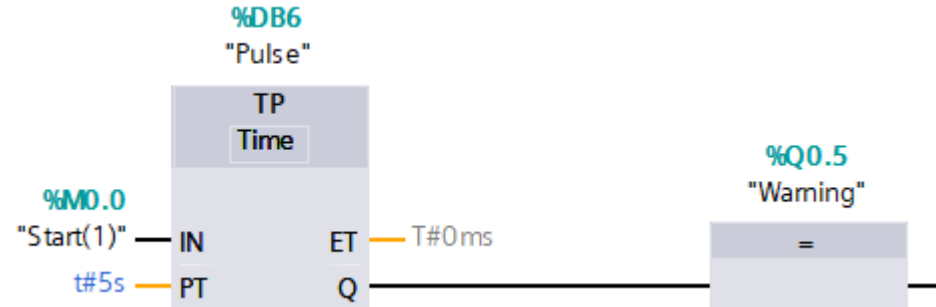


Conveyor 2 will run if conveyor 1 is on. If the control signal Q0.0 on conveyor 1 changes to 0 (i.e. the conveyor 1 is stopped), a 20-second Off Delay is triggered, after which the conveyor 2 also stops (i.e. Q0.1 mode changes to 0). If the Q0.0 status of the address changes to 1, the timer stops (and the time calculation starts again from the beginning). From the output "ET" current value of the timer can be read.

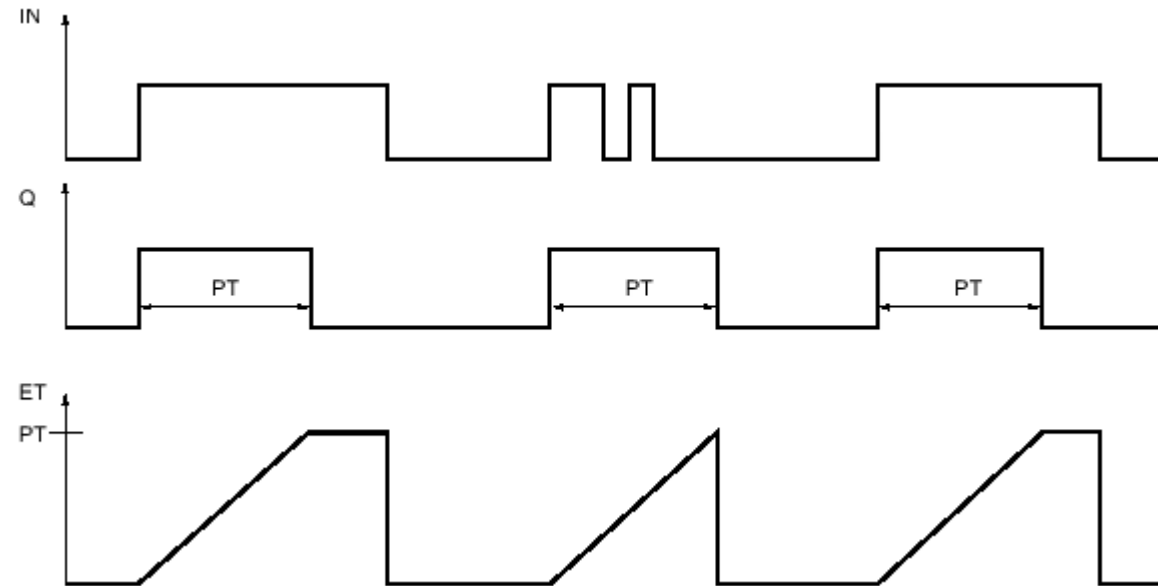
Timers

Siemens TIA Portal

Pulse



The following figure shows the pulse timing diagram of the instruction:

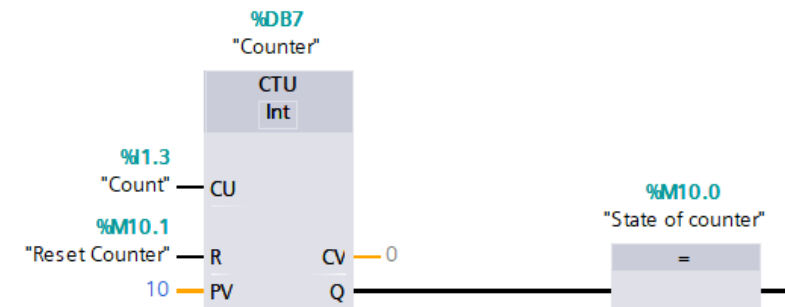


When the address M0.0 changes to mode 1, a 5-second pulse timer is triggered which starts a warning sound (Q0.5). The warning sound will turn off when the time specified in the timer has elapsed, even if the timer input is still in mode 1. From the output "ET" current value of the timer can be read.

Counters

Counters (IEC standard)

- Count Up (CTU)
 - The value of the counter increases when it detects an rising edge of the CU signal. The input R resets the counter. The value of the counter is readable from CV eg. For the comparator
- Count downs (CTD)
 - Similar, as a CTU, but goes down when detecting rising edge of CD signa.
- Count up and down (CTUD)
 - includes both count up (CU) and count down (CD) income.



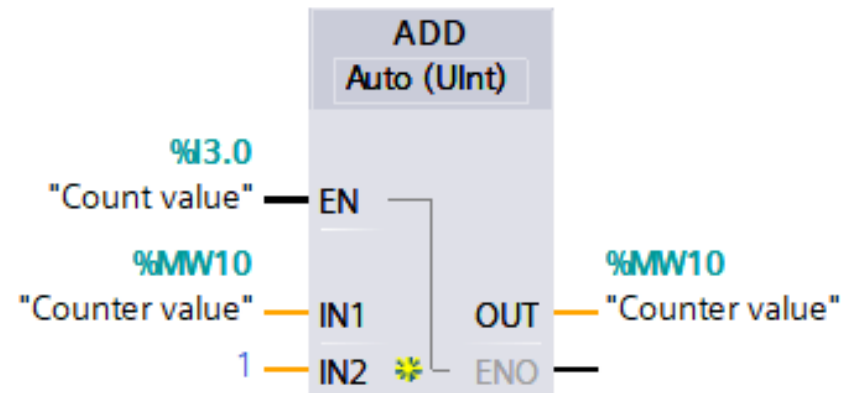
Calculation can also be done with arithmetic functions

- The counter can sometimes be replaced by a standard addition operation.
- It is worth noting, for example, the storing of information in power brakes - whether the memory area used is retentive or “not saved”. For example, number of pieces packed to the box.
- All basic calculations are available. The variables/values used must be of the same data type (data type)

Resetting MW10 could

perform e.g. with Move-operation

→ Move value 0 to the memory slot MW10



Comparators

- If the condition of comparison is true, then the departure of the comparator is also true
- In general, all the comparators familiar from the mathematics class are available: <, <=, ==, >=, > and <>
- N.B. The values to be compared must be same data type! Integer-type values can be compared to other Integer-type values, REAL-type to other REAL-typed values, etc. If necessary, you can make a type conversion (conversion policy blocks)